Heart Disease Prediction Using Machine Learning

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Executive Summary

This project develops a machine learning model to predict heart disease risk using the CDC BRFSS 2015 dataset containing 253,680 health survey responses. We implemented Random Forest and Logistic Regression models, prioritizing recall (sensitivity) over accuracy to minimize false negatives in medical diagnosis. The Random Forest model achieved 81.7% recall, successfully identifying over 4 out of 5 patients with heart disease, making it suitable for clinical screening applications.

1. Overview of the Dataset

1.1 Dataset Description

The CDC Behavioral Risk Factor Surveillance System (BRFSS) 2015 dataset is a comprehensive health survey dataset that provides critical insights into heart disease risk factors across the United States population.

Dataset Specifications:

Attribute	Description
Total Samples	253,680 responses
Features	22 health and demographic indicators
Target Variable	HeartDiseaseorAttack (binary: 0=No, 1=Yes)
Data Type	All numerical features
Missing Values	None (complete dataset)
Class Distribution	Imbalanced (9.4% positive cases)

1.2 Key Features

The dataset includes 22 features categorized into several groups:

- Health Indicators: HighBP, HighChol, CholCheck, BMI, Smoker, Stroke, Diabetes, GenHlth, PhysHlth, MentHlth, DiffWalk
- Lifestyle Factors: PhysActivity, Fruits, Veggies, HvyAlcoholConsump
- Healthcare Access: AnyHealthcare, NoDocbcCost
- Demographics: Age (13 categories: 18-24 to 80+), Sex, Income, Education

1.3 Class Imbalance Challenge

The dataset exhibits significant class imbalance with only 23,893 positive cases (9.4%) compared to 229,787 negative cases (90.6%), representing a 9.6:1 ratio. This imbalance presents a critical challenge as naive models could achieve high accuracy by simply predicting "no disease" for all cases while failing to identify actual patients at risk.

1.4 Data Quality

Initial analysis revealed 23,899 duplicate records and some invalid BMI values. After data cleaning, 229,502 unique samples remained. The dataset contains no missing values, ensuring reliable model training without requiring imputation strategies.

2. Methods and Models Used

2.1 Data Preprocessing Pipeline

Step 1: Data Cleaning

- Removed 23,899 duplicate records
- Filtered invalid BMI values (< 12 or > 80)
- Final dataset: 229,502 samples

Step 2: Feature Engineering

- Dropped Income and Education columns due to limited predictive value
- Retained 20 features for model training

Step 3: Train-Test Split

- 80-20 stratified split (183,601 training, 45,901 test)
- Stratification maintains class distribution in both sets

Step 4: Feature Scaling

- Applied StandardScaler normalization (mean=0, std=1)
- Critical for distance-based algorithms and gradient descent

Step 5: Handling Class Imbalance

To address the severe class imbalance, we employed random undersampling of the majority class. We randomly selected 18,958 healthy individuals to match the number of disease cases, creating a perfectly balanced 50-50 training set of 37,916 samples. While this approach reduces training data, it significantly improves the model's ability to learn patterns associated with heart disease.

2.2 Model Selection and Configuration

Random Forest Classifier (Primary Model)

Random Forest was selected as the primary algorithm due to its robustness, ability to handle complex non-linear relationships, and resistance to overfitting through ensemble learning.

- Number of estimators: 200 decision trees
- Maximum depth: 15 levels (prevents overfitting)
- Minimum samples split: 10 (requires at least 10 samples to split a node)
- Minimum samples leaf: 5 (ensures leaf nodes have sufficient samples)
- Parallel processing: n_jobs=-1 (uses all CPU cores)

Logistic Regression (Baseline Model)

Logistic Regression serves as a baseline linear model for comparison, providing interpretable coefficients and establishing a performance benchmark.

- Maximum iterations: 1,000 (ensures convergence)
- Default L2 regularization (prevents overfitting)
- Solver: LBFGS (efficient for small to medium datasets)

2.3 Evaluation Strategy

Given the medical context where missing a disease case has far more severe consequences than a false alarm, we prioritized recall (sensitivity) as our primary evaluation metric. This ensures the model catches as many true disease cases as possible, even at the cost of increased false positives.

- Primary Metric: Recall (minimize false negatives)
- Secondary Metrics: Precision, Accuracy, F1-score, ROC-AUC
- Confusion Matrix: Detailed breakdown of prediction types
- ROC Curve: Trade-off between true positive and false positive rates

3. Results with Key Visualizations and Performance Metrics

3.1 Random Forest Performance

Performance Metrics Summary:

Metric	Value	Interpretation
Recall (Sensitivity)	81.7%	Caught 3,871 out of 4,739 sick patients
Precision	24.3%	About 1 in 4 positive predictions is correct
Accuracy	71.9%	Overall correct predictions
F1-Score	0.375	Harmonic mean of precision and recall
ROC-AUC	0.835	Strong discrimination ability

Confusion Matrix Analysis:

Category	Count	Description
True Negatives	29,119	Correctly identified healthy individuals
False Positives	12,043	Healthy individuals incorrectly flagged (false alarms)
False Negatives	868	Sick individuals missed by the model
True Positives	3,871	Correctly identified sick individuals

Medical Interpretation:

Out of 4,739 patients with heart disease in the test set, the Random Forest model successfully identified 3,871 cases (81.7%), while missing 868 cases (18.3%). The false alarm rate among healthy individuals was 29.3%, meaning roughly 3 in 10 healthy people would be referred for additional screening.

3.2 Feature Importance Analysis

The Random Forest model identified the following top 10 most influential features:

1. 1. **Age:** Age category (strongest predictor)
2. 2. **GenHlth:** General health assessment
3. 3. **HighBP:** High blood pressure
4. 4. **BMI:** Body mass index
5. 5. **HighChol:** High cholesterol
6. 6. **PhysHlth:** Number of days physical health was not good
7. 7. **DiffWalk:** Difficulty walking or climbing stairs

8. 8. **Diabetes:** Diabetes status
9. 9. **MentHlth:** Number of days mental health was not good
10. 10. **Stroke:** History of stroke

Age emerges as the dominant predictor, consistent with medical understanding that cardiovascular risk increases significantly with age. General health status and established cardiovascular risk factors (high BP, high cholesterol) follow as expected.

3.3 Logistic Regression Performance

The Logistic Regression model achieved 78.8% recall, catching 3,735 out of 4,739 sick patients while missing 1,004 cases. Though slightly lower in recall than Random Forest, it demonstrated better precision (25.6%) and accuracy (74.2%). The ROC-AUC of 0.836 indicates comparable discrimination ability.

3.4 Model Comparison

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Random Forest	71.9%	24.3%	81.7%	0.375	0.835
Logistic Regression	74.2%	25.6%	78.8%	0.387	0.836

Key Findings:

- Random Forest achieved 2.9 percentage points higher recall (81.7% vs 78.8%), catching 136 more disease cases
- Logistic Regression showed better overall accuracy (74.2% vs 71.9%) and precision (25.6% vs 24.3%)
- Both models demonstrated nearly identical ROC-AUC scores (0.835 vs 0.836), indicating similar discrimination ability
- Random Forest is recommended for medical screening due to superior recall, prioritizing patient safety

3.5 Key Visualizations Generated

The project includes comprehensive visualizations:

- Target variable distribution (pie chart and bar chart)
- Age and BMI distributions
- Disease rate by age category (bar and line plots)
- Disease rate by key risk factors (6-panel comparison)
- Correlation heatmap of all features
- Feature importance ranking (horizontal bar chart)
- Confusion matrices for both models
- ROC curves with AUC scores

- Side-by-side model performance comparison

4. Discussion

4.1 Model Performance Interpretation

The Random Forest model's 81.7% recall represents a strong balance between catching disease cases and maintaining reasonable precision. In a medical screening context, this performance is appropriate because:

- High Recall Justification: Missing a heart disease case (false negative) can have life-threatening consequences, while false alarms (false positives) lead to additional but non-harmful testing.
- Trade-off Acceptance: The 29.3% false alarm rate is acceptable for preliminary screening, as flagged individuals would undergo confirmatory testing rather than immediate treatment.
- Clinical Utility: The model could serve as a first-line screening tool to identify high-risk individuals who would benefit from comprehensive cardiovascular assessment.

4.2 Feature Insights

The feature importance analysis reveals that the model's predictions align well with established medical knowledge. Age dominates as expected, since cardiovascular risk increases exponentially with aging. The prominence of general health status (GenHlth) suggests that subjective health perception correlates strongly with objective disease presence.

Established cardiovascular risk factors (high blood pressure, high cholesterol, diabetes) appear in the top features, validating the model's learning of medically meaningful patterns. The inclusion of functional limitations (DiffWalk) and health-related quality of life measures (PhysHlth, MentHlth) indicates that the model captures both clinical risk factors and their manifestations in daily life.

4.3 Addressing Class Imbalance

The undersampling strategy successfully addressed the severe class imbalance, enabling the model to learn patterns from disease cases rather than simply predicting the majority class. While this approach reduces the effective training data size, it significantly improved recall from baseline levels that would occur with imbalanced training.

Alternative approaches such as SMOTE (Synthetic Minority Over-sampling Technique) or class weight adjustments could be explored in future work to retain more training data while maintaining balanced learning.

4.4 Comparison: Random Forest vs Logistic Regression

Random Forest outperformed Logistic Regression on the primary metric (recall) by 2.9 percentage points, translating to 136 additional disease cases identified. This advantage stems from Random Forest's ability to model non-linear relationships and complex feature interactions that Logistic Regression cannot capture.

However, Logistic Regression demonstrated competitive performance overall, with better accuracy and similar ROC-AUC. Its simpler architecture offers advantages in interpretability and computational efficiency, making it valuable for understanding linear risk factors.

5. Conclusion

A machine learning system was developed to predict heart disease risk using the CDC BRFSS 2015 dataset.

Careful data preprocessing and handling of class imbalance improved model performance.

The **Random Forest classifier** achieved **81.7% recall**, correctly identifying over 4 out of 5 individuals with heart disease.

High recall makes the model suitable for **medical screening**, where detecting disease cases is critical.

Feature importance analysis showed that age, general health status, and known cardiovascular risk factors are key predictors.